EPR believed in the principle of locality—the physical quantities describing nature are defined at specific points in spacetime, not spread out all over the place, and they interact directly only with other quantities nearby, not at a distance. Said another way, given the speed-of-light restriction of special relativity, locality would seem to imply that nothing we can do to a particle at one location can instantaneously affect measurements we might perform on another particle very far away.

On the face of it, the fact that two widely separated particles can be entangled seems to imply that locality is violated in quantum mechanics. But EPR wanted to be a little more thorough, and establish that there wasn't some clever work-around that would make everything seem local.

They suggested the following principle: if we have a physical system in a specified state, and there is a measurement we can do on that system such that we know with 100 percent certainty what the outcome will be, we associate an *element of reality* with that measurement outcome. In classical mechanics, the position and the momentum of each particle qualify as elements of reality. In quantum mechanics, if we have a qubit in a pure spin-up state, there is an element of reality corresponding to the spin in the vertical direction, but there need not be an element of reality corresponding to the horizontal spin, as we don't know what we will get when we measure that. A "complete" theory, in the EPR formulation, is one in which every element of reality has a direct counterpart in the theory itself, and they argued that quantum mechanics couldn't be complete by this criterion.

Let's take Alice and Bob and their entangled qubits, and imagine that Alice has just measured the vertical spin of her particle, finding that it points upward. We now know that Bob will also measure spin-up, even if Bob doesn't know it himself. So by EPR's lights, there is an element of reality attached to Bob's particle, saying that the spin is up. It's not that this element of reality came into existence when Alice did her measurement, as Bob's particle is very far away, and locality says that the element of reality must be located where the particle is; it must have been there all along.

But now imagine that Alice didn't do the vertical-spin measurement at all, but instead measured the spin of her particle along the horizontal axis. Let's say she measures spin-right for the particle. The entangled quantum state we started with ensures us that Bob will get the same result that Alice did, no matter what direction she chooses to measure her spin in. So we know that Bob would also measure spin-right, and by EPR's lights there is